1. Permutations and Combinations

1.1. Factorial

The factorial for a positive integer n can be expanded as

$$n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$$

An interesting result obtained from factorials is

$$0! = 1$$

- A calculator can also be used to compute the value of a factorial.

Example 1

Compute the value of $5!$

Answer

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$$

Example 2

Compute the value of $\frac{100!}{98!}$

Answer

We can re-write the numerator as

$$100 \times 99 \times 98 \times 97 \times \cdots \times 3 \times 2 \times 1 \implies 100 \times 99 \times 98!$$

So the fraction is

$$\frac{100 \times 99 \times 98!}{98!}$$

Clearly $98!$ cancels out in the numerator and denominator, leaving us with

$$\frac{100 \times 99 \times \cancel{98!}}{\cancel{98!}} = 100 \times 99$$

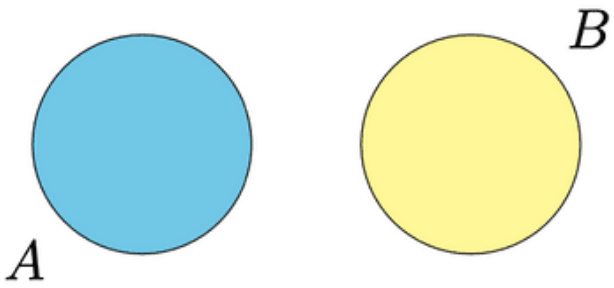
So the value we get is

$$\frac{100!}{98!} = 100 \times 99 = 9900$$

1.2. Addition and Multiplication principle

Addition Principle

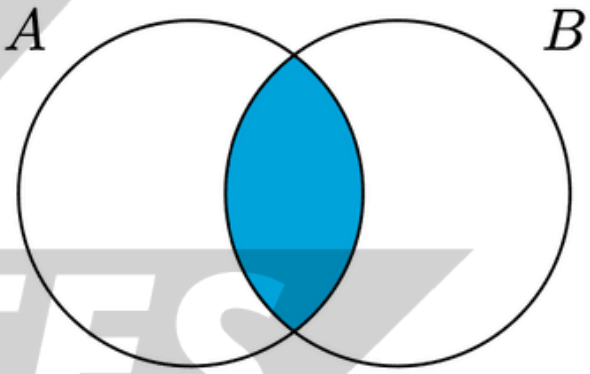
- If an event can occur in m ways and a second event, with **no common outcomes**, can occur in n ways then the first **OR** second event can occur in $m+n$ ways.



- The union of these two sets gives us the addition principle
  - $|A \cup B| = |A| + |B|$ [Where $|x|$ denotes the number of elements in that set]
- A general tip is to use the addition principle when a question uses the word "**or**"
  - To elaborate, it should be used in cases where you are finding the number of different cases which do not happen at the same time.

Multiplication Principle

- If an event can occur in m ways and a second event can occur in n ways, **after the first event has occurred**, then the two events can occur in $m \times n$ ways.



- The intersection of these two sets gives us the multiplication principle
  - $|A \times B| = |A||B|$ [Where $|x|$ denotes the number of elements in that set]
- A general tip is to use the multiplication principle when a question uses the word "**and**"
  - To elaborate, it should be used in cases where you are finding the number of different cases which happen at the same time, or after the first event has occurred.

1.3. Permutations

- A Permutation can be defined as the number of arrangements of n distinct objects where **order matters**.

  The number of arrangements of r objects from n total distinct objects can be found using

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$$^{n}P_r = \frac{n!}{(n-r)!} \quad [\text{Where } r \leq n]$$

Furthermore

$$^{n}P_n = n!$$

Example

In how many different ways can five boys be arranged in a row?

Answer

First, we begin by drawing our dashes

$$_____$$

- We have 5 possible boys to choose from to place in the first dash
  - After we place the first boy in the first dash, we have 4 possible boys to choose from to place in the second dash.
  - The same process is repeated until all the boys are arranged.

This gives us the answer as

$$5 \times 4 \times 3 \times 2 \times 1 = {}^{5}P_5 = 5! = 120$$

Example 2

In how many different ways can 3 apples be arranged in a row, from a total of 6 apples.

Answer

First, we being by drawing our dashes

$$___$$

- We have 6 possible apples to choose from to place in the first dash
  - After we place the first apple in the first dash, we have 5 apples to choose from to place in the second dash.
  - After we place the second apple in the second dash, we have 4 apples to choose from to place in the third and final dash.

This gives us the answer as

$$6 \times 5 \times 4 = {}^{6}P_3 = 120$$

1.4. Permutations with Repetitions

- As permutations only deals with the number of arrangements of **distinct** where **order matters,** we **overcount** the number of arrangements if we try to use the regular formula.

$$\text{Number of arrangements} = \frac{n!}{p! \times q! \times \cdots}$$

Where:

- n is the **total** number of objects
  - $n = p + q + \cdots$
- p is the number of p **repeated alike**, non distinct, objects
- q is the number of q **repeated alike**, non distinct, objects

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Example

Find the number of permutations of the 11 letter word MISSISSIPPI.

Answer

First we note down the number of times each letter occurs in the word.

$1 = M, 4 = I, 4 = S, 2 = P$

Using the formula we get

$$\text{Number of arrangements} = \frac{11!}{4! \times 4! \times 2!} = 34,650$$

1.5. Problem Solving Strategies

Gap Method

Let's say there are 4 girls and 4 boys. How many ways can they be seated, such that no girl is sitting next to another girl?

Answer

Let the girls be G_1, G_2, G_3, G_4 and let the boys be B_1, B_2, B_3, B_4. People are distinct, so we do not need to take in consideration any repetitions.

We can first arrange the boys in such a way that there is a gap between them, to seat the girls.

$$_\ \underline{B_1}\ _\ \underline{B_2}\ _\ \underline{B_3}\ _\ \underline{B_4}\ _$$

Observe that we have made 5 dashes, while there are only 4 girls. This is because the arrangement can be such that a girl is placed in the first dash. If a girl is not in the first dash, then the dash is empty and the boy would be considered the first in line. The same logic is used for the last dash.

We can now select the girls, that are going to be seated, using combinations. There are 5 dashes, but we only need to seat 4 girls. So out of 5 seats, we choose 4 seats to gain the total number of selections possible.

$$^{5}C_4$$

Now that we have selected 4 out of 5 seats, we can find the total number of arrangements. We will end up with $4!$ ways of arranging the 4 boys **AND** $4!$ ways of arranging the girls who will be seated. We end up with

$$^{5}C_4 \times 4! \times 4! = 2880 \text{ ways}$$

String Method

There are 3 boys and 4 girls. In how many ways can they be seated, such that all the boys are sitting in a row.

Answer

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First lets "string" together all the boy students. This string of students will be considered as **one dash**.

$$\boxed{B_1\ B_2\ B_3}\ G_1\ G_2\ G_3\ G_4$$

We can now say there will be 3! ways of arranging the boys **inside** the string amongst each other, and 5! ways of arranging the 5 total dashes. We have considered the string to be 1 dash, so there are 5! ways of arranging the 4 girl students and the string.

In total there are:

$$5! \times 3! = 720 \text{ ways of arrangement}$$

1.6. Combinations

- A combination is a way of **selecting**, or **choosing**, r objects, in **no particular order**, from n objects.

The number of ways of selecting, or choosing, r objects from n objects is:

$${}^nC_r = \frac{n!}{(n-r)! \times r!}$$

This can also be thought of as

$${}^nC_r = \frac{\text{Total number we select from!}}{\text{Number selected!} \ \times \ \text{Number not selected!}}$$

Example 1

In how many ways can 5 books and 3 magazines be selected from 8 books and 6 magazines?

Answer

- The word "and" indicates the use of the multiplication principle.

We can choose 5 books from 8 books in ${}^8C_5$ or 56 ways.

We can choose 3 magazines from 6 magazines in ${}^6C_3$ or 20 ways.

As we are trying to find the total number of selections, we multiply both results. This is the multiplication principle

Total number of ways of selection = $56 \times 20 = 1120$.

Example 2

A team of five is to be chosen from six women and five men. Find the number of possible teams in which there will be more women than men.

Answer

We can have 3 cases in which there are more women than men on the team.

$$\text{Case 1} \ - \ \underline{w}\ \underline{w}\ \underline{w}\ \underline{m}\ \underline{m} = {}^6C_3 \times {}^5C_2 = 200$$

$$\text{Case 2} \ - \ \underline{w}\ \underline{w}\ \underline{w}\ \underline{w}\ \underline{m} = {}^6C_4 \times {}^5C_1 = 75$$

$$\text{Case 3} \ - \ \underline{w}\ \underline{w}\ \underline{w}\ \underline{w}\ \underline{w} = {}^6C_5 \times {}^5C_0 = 6$$

As we cannot have all of these cases happening at the same time, due to a constraint on the number of people, we will have a choice between case 1 **OR** case 2 **OR** case 3. This is an application of the addition principle, thus we add all our cases to find the total number of selections.

$$200 + 75 + 6 = 281 \text{ ways}$$

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